RENTALREVIVE · INTERNAL TECHNICAL REFERENCE

The Rental Revenue Leak Scan

What happens while a yard is being scanned, and the exact arithmetic that produces the number at the end.

Derived from source, not documentation: `common/leaks.js`, `common/segments.js`, `frontend/src/onboard/plan.js`, `frontend/src/onboard.js`. Worked example verified by executing the live engine. Generated 6 August 2026.

Rev 9 — 6 Aug 2026: ticket brackets now follow the **iron the yard actually ticks**: every fleet entry carries a [lo, hi] window into its segment's bands (§4c), grounded in the 2026 rate pass, and the offered brackets are the union over ticked machines. Competitor radar lists every yard found (count == list), the sweep stays live while pins land, and the retired Universal Analytics row is gone. **187/187 tests green.**

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1 · What the scan is

The scan is a conversational diagnostic that a rental yard owner completes in roughly two minutes. It asks about twenty-four questions, converts the answers into a money model, and ends by showing the owner an estimated monthly recoverable figure — *together with the arithmetic that produced it*.

> **The governing design rule**
>
> Every number shown to the owner is accompanied by the formula that made it. This is deliberate and load-bearing. The buyer is a skeptical operator who has been sold "leads" before; a figure they cannot audit reads as a sales trick. So the rules are: conservative assumptions stated out loud, one formula per leak rendered as a readable string, and nothing labelled a guarantee — everything labelled an estimate.

The model rests on a single formula from the founder's audit spec (`03-MISSED-RENTAL-AUDIT §6`):

```
potential = missed opportunities  ×  close rate  ×  average job value
```

Everything below is that formula applied five times, once per leak, with different definitions of "missed opportunity" — and nothing else. There is no second model.

2 · What happens during a scan, in order

1 · The gate

The owner types a business name. If a Google Maps key is configured, Places Autocomplete returns live predictions and the yard is "locked on" — pulling name, address, coordinates, rating, review count, photos and opening hours. With no key the gate degrades gracefully: the owner types a name and the scan proceeds identically. No question in the money model depends on Google.

2 · Enrichment (parallel, non-blocking)

Behind the conversation, the Google footprint card scores the yard's public listing against a checklist — name, address, phone, website, hours, categories, photo count, star rating, review volume — and a website crawl looks for tracking signals (GA4, GTM, Meta Pixel, Google Ads). This is *observation*, not measurement, and it never enters the leak arithmetic.

3 · Act 1 — the yard

Counter phone number, machinery segments (multi-select from eight), primary segment if more than one is picked, fleet composition, fleet size.

4 · Act 2 — the market

A competitor radar sweeps the radius for rival yards (skipped without Maps). The owner names who takes jobs from them, and why those rivals win.

5 · Act 3 — the numbers

Monthly inquiry volume, inbound channels, marketing posture, average job value, close rate, counter headcount.

The meter arms here

— until an average job value exists there is no money model and every leak reads zero.

6 · Act 4 — the five leaks

Missed calls, after-hours handling, quote speed, standing quote pile, lapsed accounts, outbound posture. Each answer immediately re-prices its leak and the live meter climbs.

7 · Cooking → reveal → solution → report

A processing overlay, then the headline number, then the recommended offer and both paths, then the full arithmetic table. Finally the send-form close.

Live recalculation

The engine is safe to call at any point. Unanswered questions contribute exactly zero, which is what lets the meter climb as the conversation goes on rather than appearing at the end. If the owner taps any previous answer to change it, `refreshVerdict()` tears down the reveal, solution and report and rebuilds all three from scratch — the whole model re-runs, plan included. There is no incremental patching of a stale figure.

3 · How answers become numbers

The owner never types a number. Every quantitative question is a band, and each band maps to a single defensible midpoint. This is the first place estimation enters the model, and it is intentional: operators cannot recall exact figures, but they can reliably place themselves in a range.

Monthly inquiries

ANSWER	VALUE
Under 25	18
25 – 60	42
60 – 120	88
120+	165

Missed calls per week RANGES + EXACT

Five ranges for the owner who has never counted, plus an "I know my exact number" entry (0–150) for the one who has. Ranges commit their midpoint as a *number* — no new strings enter the probe vocabulary.

OPTION	PRICED AS
Almost none	0
1 – 5 a week	3
6 – 15 a week	10
16 – 40 a week	26
More than 40	55
I know my exact number	as typed, ≤150

The 16–40 and 40+ ranges exist because benchmarks put a mid-size shop at 40–90 missed a week. The probe still substitutes the legacy band strings (3 / 10 / 22), see §4b.

Average ticket RANGES + EXACT

The segment's own four ticket bands as tap options (committing their midpoints as numbers), plus an "I know my exact number" entry for the owner who can read it off last month's invoices — see §4c.

OPTION (CRANES)	PRICED AS
Under $2,000	$1,400
$2,000 – $5,000	$3,300
$5,000 – $12,000	$8,000
$12,000+	$18,000

Lapsed accounts

ANSWER	VALUE
Just a few	5
10 – 25	17
25 – 75	48
75+	115

Quote speed → share lost to lag

ANSWER	LOSS
Inside the hour	2%
Same day	7%
Next day	15%
Two days or more	25%

After-hours handling → weight

ANSWER	WEIGHT
Nothing, it just rings	1.15
Voicemail	1.00
Answering service	0.60
Someone on call	0.35

I know my exact number	as typed, ≤ $54,000

Standing quote pile

ANSWER	VALUE
Under 20	12
20 – 50	34
50 – 150	95
150+	220

Outbound posture → jobs missed per month

ANSWER	JOBS/MO NEVER QUOTED
Yes, we work them	0
Now and then	1.2
No, we wait for the phone	2.5

Held deliberately small. This is the softest input in the model — it estimates demand the yard has never observed, so it is the one most open to challenge.

4b · Missed calls: why it became a slider

Missed calls was originally four bands topping out at "15+ a week", priced at 2 / 6.5 / 14. Two things were wrong with that: the numbers were far too low, and the top band collapsed every busy yard into a single bucket.

What the published benchmarks say

MEASURE	BENCHMARK
Missed share of inbound volume, normal week	20 – 30% (27% is the 2024–26 rolling average across the trades)
Missed share at seasonal peak	40 – 50%
Mid-size shop, 200–300 calls/week	40 – 90 missed calls weekly
Typical contractor described as "losing real money"	5 – 10 missed calls/week
Inbound arriving outside counter hours	25 – 40%

A yard answering "a few a week" is describing 5–10 calls, not 2 — and a mid-size shop is nowhere near the old 14-a-week ceiling. The question is now **five ranges plus an exact-number entry**: ranges for the owner who has never counted (each commits its midpoint as a number), and a typed figure (0–150) for the one who checks a call log. Either way the engine prices a number directly.

> **Two input shapes, one price**
>
> The owner's answer is now a **number**. The AD-11 probe cannot produce one — it measures a *rate* (share of test calls that never reached a human), not a weekly volume — so it still substitutes one of the band **strings**, which

resolve through the re-cut midpoints below. The engine accepts both, and the numeric score thresholds are the band boundaries exactly (1–5 → 2 pts, 6–15 → 4, 16+ → 5) so the two agree at the seams.

RETAINED BAND (PROBE PATH ONLY)	PRICED AT	WAS
Almost none	0	0
1 – 5 a week	3	2
6 – 15 a week	10	6.5
15+ a week	22	14

The structural flaw this exposed

Missed calls is the only leak in the model whose input is an **absolute count** rather than a share of the yard's own flow. Every other leak scales with inquiry volume or ticket value; this one did not. Two yards — one taking 18 inquiries a month, one taking 165 — who both answered the same band were priced identically.

That was survivable with a top band of 14. With realistic bands it is not: a yard reporting 42 inquiries a month would be priced on 95 missed rentals. So the revision adds a bound.

The volume bound

```
missedRaw     = missed/week × 4.33 × afterHoursWeight
missedMonthly = min( missedRaw , inquiries/month )
```

A yard cannot plausibly be missing more new-rental calls in a month than actually reach it. The cap sits at 100% of observed inquiry volume — equivalent to a 50% miss rate, the top of the published peak range — so it binds only the implausible cases. When it fires, it is printed in the owner's arithmetic rather than silently shrinking the number.

Known consequence: the cap flattens the top of the slider

For a low-volume yard the cap binds well before 100, so dragging past that point stops changing the money (the 0–25 leak score still climbs). A yard reporting 42 inquiries a month prices the same at 18 a week as at 100. That is an honest ceiling — you cannot miss more new-rental calls than reach you — but it does mean the slider's upper half is inert for small yards. Worth revisiting if real distributions ever land.

4c · Average ticket: the same treatment

Ticket had the same defect as missed calls, and it mattered more: **ticket multiplies every one of the five leaks**, so bucketing it was the coarsest single decision in the model. Four bands forced a $12,000 lift and a $60,000 plant relocation into one "$12,000+" bucket priced at $18,000 — a 3× misprice at the top of the range.

It is now the segment's own four ticket bands as tap options — each committing its *midpoint as a number* — plus an exact-invoice entry for the owner who knows. Committing numbers rather than band labels also closes a cross-segment trap: a stored label like "Under $2,000" stops resolving if the owner later changes their primary segment, while a number stays a number.

The typed entry is capped at twice the old slider ceiling (≈ 3× the top band midpoint), derived from the segment's own bands so its economics stay defined in one place:

```
cap = round( dearest midpoint × 1.5 ) × 2      ← $54,000 for cranes, $180,000 for heavy haul
```

SEGMENT	BAND MIDPOINTS	EXACT-ENTRY CAP
Compact & Small Machinery	$340 – $6,000	$18,000
Cranes & Lifting	$1,400 – $18,000	$54,000
Heavy Haul, Rigging & Specialty	$3,200 – $60,000	$180,000

Fleet windows — the brackets follow the iron

Each fleet entry now carries a `[lo, hi]` window into its segment's four bands (`fleetBands` in `segments.js`), grounded in the same rate data as the bands themselves. The brackets offered are the union over every machine the yard ticked:

CRANES & LIFTING EXAMPLE	WINDOW	WHY
Carry deck & industrial · Rigging & lift gear	`bands 0–1`	Small-ticket in-plant work
Mobile / all-terrain · Operated crews	`bands 1–3`	$1,200–2,500/day + crew + transport
Crawler cranes	`bands 2–3`	Big iron, long jobs
Tower cranes	`band 3`	Monthly installs, $10k+/mo before labor

A yard renting only carry decks and rigging sees two brackets, not a "$12,000+" option it can never mean. Guard rails: a machine with no window falls back to its segment's full bands (narrowing happens on good data, never on missing data), the custom box takes any number so a hidden bracket cannot block a true answer, and two tests enforce the map — every fleet string has a valid window, no orphan keys, and every segment's cheapest and dearest bracket is reachable by some machine.

A typed ticket above the cap commits the cap with a "+" shown in the transcript pill — and the §4e plausibility bound still governs the total downstream, so even a $54,000 ticket cannot produce a total exceeding half the implied bookings.

No silent fallback

An unanswered or unrecognised ticket resolves to `0`, and `live` keys off `ticket > 0` — so the entire money model stays at zero rather than pricing off a default the owner never gave. This is asserted in the test suite for empty string, null, undefined, 0, negative, NaN and unknown labels.

Still bands — and arguably shouldn't be

Three more inputs are absolute quantities carrying the same top-band collapse. They were left as bands in this pass:

INPUT	TOP BAND	PRICED AT	CONSEQUENCE
Monthly inquiries	120+	165	Also the ceiling for the missed-call volume bound (§4b)
Standing quote pile	150+	220	Drives the desk gate and the 60-day sequence counts
Lapsed accounts	75+	115	Smallest leak; lowest impact of the three

4 · The five recovery assumptions

These are described in the source as "*the only invented numbers in the model*". They are held low on purpose and are printed verbatim at the bottom of every report.

ASSUMPTION	VALUE	MEANING
`reachable`	45%	Of missed callers, the share still reachable after the fact
`winnable`	50%	Of quotes lost to response lag, the share speed would have saved
`revivable`	8%	Of a cold quote pile, the share that comes back with follow-up
`reactivated`	12%	Of lapsed accounts, the share that rents again when worked
`quoteRate`	80%	Of inquiries, the share that actually becomes a quote

Two derived quantities used throughout

```
quotes/month   = inquiries × 0.80
close          = the owner's stated close rate, or the segment default
WEEKS          = 4.33   (months per week conversion)
```

If the owner does not set a close rate, the segment's `defaultClose` is used — ranging from 25% for Heavy Haul & Specialty to 50% for Compact & Small Machinery.

4d · Market grounding — where each constant now sits

Rev 5 replaced "held low on purpose" with "held at or below the low end of the published range". The values barely moved; what changed is that each one now has a benchmark to answer to.

CONSTANT	VALUE	PUBLISHED BENCHMARK	WHERE IT SITS
`reachable`	45%	45% of voicemail-leavers have already called a competitor (≈55% still in play); manual callbacks convert <30%; sub-minute automated text-back recovers 93%	Just under the still-in-play ceiling, for a prompt worked callback
`winnable`	50%	35–50% of sales go to the first responder; 78% of buyers pick whoever answers first; sub-5-minute response converts 21% vs 2.3% after a day	Top of the first-responder share, well below the 78%
`revivable`	8%	B2B dormant-lead reactivation recovers 5–15%; 80% of deals need 5+ touches, 92% of reps quit by 4	Low-middle of the range
`reactivated`	12%	B2B lapsed-customer win-back 5–15% (email-only 8–15%, full campaigns 10–30%)	Mid-range of the most conservative bracket
`quoteRate`	80%	—	Operational assumption; not benchmarked

The lag-loss curve, re-cut

ANSWER	WAS	NOW	ANCHOR
Inside the hour	2%	2%	Residual — speed is not this yard's leak
Same day	6%	7%	Close lifts ~2.6× (12% → 32%) moving from the 24h bucket to minutes; 35–50% of sales go to the first responder. The two-day band is held at half the 35% floor — a slow yard loses plenty, but not all of it to lag.
Next day	13%	15%	

Two days or more	21%	25%

Second pass — the scan-side constants

CONSTANT	VALUE	PUBLISHED BENCHMARK	VERDICT
Segment close defaults	25–50%	Rental-industry inquiry→confirmed runs 38–48% (top quartile 50%+, sub-30% flags coverage). Compounded with the 80% quote rate, our defaults imply 20–40% inquiry→booked	At or below the band for every segment — conservative by construction
Crane ticket bands	$1.4k–18k	Operated mobile crane $1,200–2,500/day mid-size + operator + transport; 1–3 day lifts land in the $2,000–5,000 band	Held
Earthmoving bands	$1.3k–16k	Excavator averages $719/day · $2,021/wk · $5,108/mo	Held — weekly/monthly bands bracket the published rates
Aerial bands	$520–8k	Scissor $495/wk · $984/mo; boom $1,050–1,950/wk · $2,700–4,600/mo	Held
Footprint: photos ≥ 10 / 20	10 / 20	10+ photos ≈ 2× engagement; 20+ ≈ +18% clicks; the average verified profile has under one photo; home-services profiles average ~70	Held
Footprint: reviews ≥ 25	25	59% of consumers only trust a rating backed by 20+ reviews; average local business ~39; expectation of reliability ~40	Held — above the trust floor, below the average
Footprint: rating ≥ 4.0	4.0	Sub-4.0 ratings are discounted outright; most-trusted band 4.2–4.5	Held — the published trust floor exactly

Sources: Gitnux, Aloware and LeadResponse speed-to-lead compilations; Dialfyne / Sawy / SchedulingKit missed-call studies; UnboundB2B and Cleverly win-back benchmarks; ZoomInfo and Invesp follow-up statistics; PulseRevOps rental-industry KPI benchmarks (the one machinery-rental-specific source found); DOZR and BigRentz 2026 rate guides; BrightLocal, Birdeye and WebFX Google Business Profile studies. Compiled 6 Aug 2026. The recovery-rate figures remain cross-industry service-trade benchmarks; the close-rate band and rate cards are rental-industry-specific. Re-cut against the yard's own probe data when it lands (§4b).

4e · The plausibility bound

Each leak is individually conservative, but five of them compound against the same flow — and with the worst answer everywhere, the sum exceeded what the yard's own answers imply it *books*. The fix is a bound on the total:

```
implied bookings/mo = quotes × close × ticket
reported total      = min( sum of leaks , 50% × implied bookings )
```

Claiming leak = 50% of bookings already means claiming a third of the yard's entire won-plus-lost flow is recoverable. Past that line, the honest reading is "the answers are noisy", not "the leak is bigger".

The bound is a ledger line, not a silent shrink

When it binds, the report table gains a row: "*The five leaks sum to $1,000,204 — but your own answers imply ≈ $1,069,200/mo in booked work (132 quotes × 30% booked × $27,000), and we will not claim you are leaking more than half of what you book*" with the deduction in the amount column — so the rows still sum to the headline. Same rule as the missed-call volume cap: a smaller number the owner cannot audit is worse than a big one.

Three deliberate edges: the bound needs a revenue base, so it cannot exist until inquiries are answered — the early one-leak-at-a-time meter climb is never touched. The 0–25 leak score is *not* clamped — urgency is not money, and a yard with absurd answers still deserves a 25/25. And the self-managed path's recovery subtotal is bounded by the reported total, so the cheap path can never appear to recover more than the whole leak.

5 · The five leak formulas

Each leak is computed independently and contributes zero until its question is answered. If no average job value has been given, `live` is false and **every leak is zero** regardless of other answers — there is no money model without a ticket.

Leak 1 — Missed calls

Calls that ring out during lunch, after hours, or while everyone is loading a truck.

```
missedRaw     = missed/week × 4.33 × afterHoursWeight
missedMonthly = min( missedRaw , inquiries/month )      ← volume bound, §4b
callLeak      = missedMonthly × 45% reachable × close × ticket
```

The after-hours weight is the only compounding modifier in the model. A yard where the phone simply rings out is penalised 15%; one with someone genuinely on call has its missed-call leak cut by 65%. The volume bound was added 6 Aug 2026 alongside the re-cut bands — see §4b.

Leak 2 — Slow quotes

Quotes that died because the number arrived too late, not because of price.

```
speedLeak = quotes/mo × lagLoss × 50% winnable × ticket
```

Note the deliberate honesty here: a yard answering inside the hour still loses quotes, just not *to lag*. The 2% floor prevents the model from claiming perfect speed recovers everything.

Leak 3 — Quotes going cold TWO NUMBERS

This is the only leak that produces both a recurring and a one-time figure, and the distinction matters commercially.

```
coldPerMonth  = quotes/mo × (1 − close)
pileFlowLeak  = coldPerMonth × 8% revivable × ticket        ← recurring, monthly
pileStanding  = pileCount    × 8% revivable × ticket        ← one-time, today
```

The flow is what goes cold every month and enters the monthly total. **The standing pile** is what is sitting on the shelf right now — reported separately and never added to the monthly figure. Conflating the two would double-count.

Leak 4 — Quiet accounts

Contractors who used to rent and simply stopped, with no complaint and no goodbye.

```
quietLeak = ( lapsedCount × 12% reactivated × ticket ) / 12 months
```

The division by twelve is the most conservative single decision in the engine. Reactivating a lapsed account is treated as producing *one* job spread across a whole year, not a recurring monthly rental. This is why quiet accounts almost always price as the smallest leak even when the raw count is large.

Leak 5 — Jobs you never hear about

Work happening inside the radius that the yard is never invited to quote.

```
outboundLeak = missedProjects/mo × close × ticket
```

The only leak with no recovery-rate discount — because the model already discounts it at source by holding `missedProjects` to at most 2.5 jobs a month.

Aggregation

```
rawMonthly   = sum of all five leak amounts
implied      = quotes/mo × close × ticket          ← what the answers say the yard books
monthly      = min( rawMonthly , 50% × implied )   ← the plausibility bound, §4e
annual       = monthly × 12
pileStanding = reported separately, never added to monthly
dominant     = the single largest leak by amount — drives the headline and the offer
```

6 · The leak score, 0–25

Running alongside the money model is a wholly separate urgency score. It uses the same answers but **ignores ticket value entirely —** a small yard with a broken quote path can score higher than a large one with a tidy counter. Five leaks, 0–5 each.

MISSED CALLS /WK	PTS	QUOTE SPEED	PTS	QUOTE PILE	PTS
Almost none	0	Inside the hour	0	Under 20	1
1 – 5	2	Same day	2	20 – 50	3
6 – 15	4	Next day	4	50 – 150	4
16+	5	Two days or more	5	150+	5

QUIET ACCOUNTS	PTS	OUTBOUND	PTS	AFTER-HOURS MODIFIER (CALLS ONLY)	ADJ
Just a few	1	Yes, we work them	0	Nothing / Voicemail	0
10 – 25	3	Now and then	3	Answering service	–1
25 – 75	4	No, we wait for the phone	5	Someone on call	–2
75+	5			*Clamped to 1–5 if any calls are missed*	

The four bands

SCORE	BAND	WHAT IT IMPLIES
0 – 7	**Low urgency**	Counter is holding. Explicitly do not sell a managed service.
8 – 14	**Starter fit**	Real leak, but system-shaped — the team can run it internally.
15 – 20	**Managed sprint fit**	More than half the leak is work nobody has time to do.
21 – 25	**Desk candidate**	Only if the six-point qualification gate also agrees.

7 · Segment parameters

Eight segments each carry their own ticket bands and default close rate. The same answers in different segments produce very different money.

SEGMENT	DEFAULT CLOSE	TICKET BAND MIDPOINTS (USD)
Cranes & Lifting	30%	1,400 · 3,300 · 8,000 · 18,000
Earthmoving & Excavation	35%	1,300 · 3,300 · 7,800 · 16,000
Aerial & Access	45%	520 · 1,300 · 3,200 · 8,000
Compact & Small Machinery	50%	340 · 950 · 2,500 · 6,000
Material Handling & Forklifts	40%	700 · 1,900 · 5,200 · 13,000
Road, Concrete & Compaction	35%	1,000 · 2,600 · 6,800 · 16,000
Power, Climate & Site Services	45%	700 · 2,100 · 6,400 · 18,000

Heavy Haul, Rigging & Specialty	25%	3,200 · 9,500 · 25,000 · 60,000

Segment also controls vocabulary throughout the transcript — a crane yard is asked about "lifts", a forklift dealer about "rentals" — and supplies the narration used to frame each leak.

8 · From diagnosis to answer

A number alone is a diagnosis, not an answer. The solution engine converts the priced leaks into a recommendation, and it does so from documented exclusion lists rather than sales preference.

Which path can close which leak

The self-managed starter explicitly excludes missed-call management, human quote follow-up, and customer reactivation. Three of the five leaks are therefore *structurally* uncloseable by the cheap path — so the engine can price exactly what each path leaves on the table.

LEAK	DOCUMENTED FIX	LIVE	SELF-MANAGED?
Missed calls	Missed inquiry workflow	Week 1	NO
Slow quotes	QuickyQuotes intake on your site	Week 1	YES
Quotes going cold	Worked quote pile, three touches	Week 2	NO
Quiet accounts	Account reactivation	Week 4	NO
Jobs you never hear about	Project Radar signals	Week 3	YES

The fastest fix

Not the biggest leak — the biggest leak that is **live in week one**. Chasing a pile often pays more, but it cannot start until the intake stops refilling it, so week-one fixes win the recommendation.

The Full Revenue Desk gate

The premium offer is never named unless at least **three of six** criteria are met. The gate is shown to the owner, met or unmet — a gate you can see is a gate you can trust.

#	CRITERION	TEST
1	100+ inbound inquiries a month	`inquiries ≥ 100`
2	50+ unclosed quotes available to work	`pileCount ≥ 50`
3	Evidence of missed calls or slow response	`missedCalls ≠ none OR speed ≥ next day`
4	Follow-up capacity genuinely short	`pileCount ≥ 20 AND team ≠ 4+`
5	Average job value supports the fee	`ticket ≥ $2,500`
6	Consistent volume across the operation	`always unknown — human call only`

Criterion 6 can never be satisfied by a scan. The maximum any scan can reach is five of six, and this is deliberate: the model refuses to fully qualify a premium sale without a human conversation.

The recommendation ladder

```
score ≤ 7                    → Revenue Leak Audit Pack     $297 one time
score ≤ 14                   → Pipeline Starter            $397 + $197/mo
score ≤ 20  OR  gate fails   → 60 Day Revenue Recovery Sprint   $2,000/mo × 2
score ≥ 21  AND gate passes  → Full Revenue Desk           $5,000 – 6,000/mo
```

The `OR gate fails` clause is the important one. A yard can score 24/25 — unambiguous desk territory — and still be recommended the sprint, with the reason stated plainly: "*Selling you the desk today would be selling ahead of the proof.*" The score alone can never trigger the most expensive offer.

The close is never yes/no

The scan ends with two paths priced side by side, not a single ask: *would you rather your team operate the system internally, or would you rather we manage the recovery for 60 days?* Both options carry their real exclusions in writing. And before either, a free real-world verification is offered — three inbound calls at different times, one timed quote request, scored out of ten. If the counter scores 8+, the owner is told so and there is nothing to sell.

9 · Worked example

The following was produced by executing the live engine, not by hand. Inputs are a mid-sized crane yard.

Inputs — Cranes & Lifting · ticket band $2,000–5,000 (midpoint **$3,300**) · close rate **40%** · **42** inquiries/mo · **3** missed calls/week (band “1 – 5 a week”) · after hours to **voicemail** · quotes out **same day** · **34** open quotes · **17** lapsed accounts · outbound worked **now and then** · **2–3** at the counter.

Derived: quotes/month = 42 × 0.80 = **33.6**. Cold per month = 33.6 × (1 − 0.40) = **20.2**.

LEAK	THE MATH	PER MONTH
Missed calls	`3/wk × 4.33 × 45% × 40% × $3,300`	$7,716
Slow quotes	`33.6 × 7% × 50% × $3,300`	$3,881
Quotes going cold	`20.2 × 8% × $3,300`	$5,322
Quiet accounts	`(17 × 12% × $3,300) / 12`	$561
Jobs never heard about	`1.2 × 40% × $3,300`	$1,584
Estimated recoverable	**`— per month`**	**$19,064**

Annual: $228,769. **Standing pile** (separate, one-time): 34 × 8% × $3,300 = **$8,976**.

What the engine concluded

Leak score	13 / 25 → **Starter fit**
Dominant leak	Missed calls — $7,716/mo CHANGED
Fastest fix	Missed calls → Missed inquiry workflow (Week 1)
Desk gate	3 of 6 met (criteria 3, 4, 5) — passes, but score keeps it at starter
Recommendation	Crane Pipeline Starter — $397 activation + $197/mo

Read the two paths against each other

PATH	RECOVERS	LEAVES OPEN	CLOSES
Self-managed starter	$5,465	$13,599	Slow quotes, Jobs never heard about
Managed sprint	$19,064	$0	All five

The cheap path recovers 29% of the identified leak. That gap is not a rhetorical device — it falls directly out of the starter's documented exclusions, and it is shown to the owner in dollars.

Note the deliberate tension the engine does not hide: it recommends the starter here, even though the sprint recovers three and a half times more, because a 13/25 score does not justify a managed service. The recommendation follows the score, not the revenue.

10 · Known limitations

The upper sanity bound — RESOLVED in Rev 6

This section previously tracked a runaway ceiling: worst-band answers produced $586k, then $683k, then $865k a month as each grounding pass made the inputs more realistic — every version dwarfing what the same answers implied the yard books. The plausibility bound (§4e) closes it. The same worst-case inputs now report:

RAW SUM OF LEAKS	$865,206/MO	IMPLIED BOOKINGS	$712,800/MO
Reported (bounded)	**$356,400/mo**	As share of won+lost flow	33%

$4.28M a year is still a huge number — but it is now half of what a yard reporting 165 inquiries at $18,000 average tickets books, rather than 120% of it. The ceiling scales with the yard instead of dwarfing it, and the deduction is printed in the owner's own ledger. The project's `CONTEXT.md` open-threads entry for this item can be closed.

Other open items

Footprint thresholds are guesses. The photo, review and rating bars in the Google footprint scorecard were picked by feel rather than from a distribution.
The outbound input is the softest. It estimates demand the yard has never observed, so it is the least defensible number if challenged.
The landing page and the scan sell different things. The scan is on the corrected offer ladder; the marketing site still shows retired names and invented pricing. That divergence is known and needs a founder decision.

11 · Why it is built this way

Three principles are enforced in code rather than left to discipline:

Auditable arithmetic. Every figure ships with its formula. The report closes by restating all four recovery assumptions as percentages and inviting the owner to change any answer and watch the model re-run.

Conservative by construction. Recovery rates are held low. Lapsed accounts are divided across twelve months. The standing pile is quarantined from the monthly total. Outbound is capped at 2.5 jobs. Each choice reduces the headline number.

No measurement without a measurement. An unconfigured channel is reported as unmeasured, never simulated. The scan degrades to "we could not check this" rather than inventing a result — the failure mode that has already shipped as a bug in two sibling products.

Every figure in this document was read from source or produced by executing the engine. Recovery assumptions, band midpoints and scoring tables reflect the code as it stands at commit `de76e34`.